

Generative Artificial Intelligence for Hyperspectral Sensor Data: A Review

Diaa Addeen Abuhani , *Member, IEEE*, Imran Zualkernan , *Member, IEEE*, Raghad Aldamani ,
and Mohamed Alshafai , *Member, IEEE*

Abstract—Airborne platforms and satellites provide rich sensor data in the form of hyperspectral images (HSI), which are crucial for numerous vision-related tasks, such as feature extraction, image enhancement, and data synthesis. This article reviews the contextual importance and applications of generative artificial intelligence (GAI) in the advancement of HSI processing. GAI methods address the inherent challenges of HSI data, such as high dimensionality, noise, and the need to preserve spectral-spatial correlations, rendering them indispensable for modern HSI analysis. Generative neural networks, including generative adversarial networks and denoising diffusion probabilistic models, are highlighted for their superior performance in classification, segmentation, and object identification tasks, often surpassing traditional approaches, such as U-Nets, autoencoders, and deep convolutional neural networks. Diffusion models showed competitive performance in tasks, such as feature extraction and image resolution enhancement, particularly in terms of inference time and computational cost. Transformer architectures combined with attention mechanisms further improved the accuracy of generative methods, particularly for preserving spectral and spatial information in tasks, such as image translation, data augmentation, and data synthesis. Despite these advancements, challenges remain, particularly in developing computationally efficient models for super-resolution and data synthesis. In addition, novel evaluation metrics tailored to the complex nature of HSI data are needed. This review underscores the potential of GAI in addressing these challenges while presenting its current strengths, limitations, and future research directions.

Index Terms—Diffusion models, generative adversarial networks (GANs), generative artificial intelligence (GAI), generative neural networks (GNNs), hyperspectral images.

I. INTRODUCTION

SPECTRAL images are images captured through a continuous waveband. While RGB images are theoretically spectral images [1], as they capture light in three broad spectral bands (red, green, and blue), the term *spectral* is typically used to describe images with more than three bands where the light hitting each pixel is further broken down into many other spectral bands. These additional bands provide more in-depth information about the imaged object [2]. Spectral images can be divided

into two categories based on the number of wavebands captured: multispectral images (MSIs) and hyperspectral images. A MSI typically focuses on a specific portion of the spectrum that ranges between 4 and 10 discrete wavebands [3]. An image can be considered a HSI when the number of wavebands extends to tens or hundreds. HSIs are usually more informative than MSIs because having many spectral bands allows for more detailed discrimination of features and reduce uncertainty. The multidimensional HSIs provide a large amount of spectral and spatial information that can be highly beneficial in building machine learning applications including domains, such as remote sensing [4], medical imaging [5], [6], and security [7]. However, building these machine learning applications often presents challenges. Hyperspectral images (HSIs) are characterized by their high dimensionality, capturing data across numerous spectral bands, which results in substantial data volumes and significant computational requirements for storage, processing, and analysis [8]. A further challenge lies in the limited availability of open-source datasets, most of which are geographically concentrated in regions like North America and Europe. This lack of diversity, often referred to as domain shifting, restricts the ability of models to generalize across different environments [9]. In addition, many datasets suffer from low spatial resolution—a tradeoff between capturing detailed spectral information and maintaining spatial precision—which complicates tasks, such as classification and segmentation [10]. The quality of labels within datasets is another critical issue. Noisy labels, often stemming from poor acquisition conditions, environmental interference, or inadequate preprocessing, undermine the reliability of models trained on such data [11]. Further complicating these issues are problems related to sensor performance. Dead pixels and random intensity spikes frequently arise due to electronic malfunctions, environmental factors, or imperfections in sensor design. While minor anomalies can often be mitigated through preprocessing, severe issues can render portions of the data unusable [12]. These challenges collectively highlight the complexity of working with HSIs and underscore the need for robust methodologies to address them. Table I summarizes the current challenges in HSI applications for machine learning purposes. Challenges, such as low spatial/temporal resolutions, noisy labels, and dead pixels are more common in satellite applications but could occur in other imaging platforms.

Generative neural networks (GNNs), initially designed for processing RGB images, have shown significant potential in addressing the unique challenges posed by HSIs. Techniques,

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The authors are with the Department of Computer Science and Engineering, American University of Sharjah, Sharjah 26666, UAE (e-mail: b00086137@aus.edu; izualkernan@aus.edu; g00085944@aus.edu; b00087311@aus.edu).

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TABLE I
CHALLENGES IN HYPERSPECTRAL IMAGING

Challenge	Description
High dimensionality	Hyperspectral images capture information across numerous bands in the electromagnetic spectrum, leading to large data volumes. While the analysis is not inherently complex, the high dimensionality demands significant computational resources and time, posing practical challenges for processing and analysis efficiently.
Domain shifting	Available hyperspectral datasets are very limited in terms of geographical location, mostly captured over North America, making it challenging to create geographically scalable solutions.
LR	Hyperspectral images are captured using satellites with LR sensors, which complicates accurate classification and segmentation of various
Noisy labels	Noisy labels are common in hyperspectral images, often due to poor preprocessing or sensor malfunctions, leading to inaccuracies in data labeling.
Dead pixels	Dead pixels and random spikes result from instrumental malfunctions, causing empty pixels within hyperspectral images.

such as latent diffusion models (LDMs) have been employed to manage the high dimensionality and computational complexity of HSI data [13], [14]. Similarly, super-resolution generative adversarial networks (SRGANs) have been applied to improve the spatial resolution of HSIs, which often suffer from low-resolution (LR) sensors [15]. In addition, GAN-based architectures have been utilized to generate synthetic HSI data, effectively addressing the issue of data imbalance in this domain. However, adapting GNNs to HSIs introduces additional challenges, including mode collapse, training instability, and non-convergence [16]. These problems are further exacerbated by the spectral complexity and high-dimensional nature of HSIs, which demand innovative approaches to preserve spectral-spatial correlations while ensuring computational efficiency.

This article provides a comprehensive review of current developments in using generative models for machine learning applications with HSIs, focusing on key tasks, such as feature extraction, image resolution enhancement, and data synthesis and translation. We shed light on the most common techniques to adapt different generative architectures to spectral images followed by a thorough discussion of the current trends and possible research gaps.

II. BACKGROUND

This section covers the background of HSI data acquisition processes, common HSI datasets, and evaluation metrics used to evaluate machine learning applications using HSI.

A. HSI Data Acquisition

HSIs are captured through specialized sensors that detect a wide range of wavelengths. Typically, the decision of which sensors to use is based on application requirements. For example, airborne platforms, where sensors are mounted on aircraft, are

commonly used. Airborne visible infrared imaging spectrometer (AVIRIS) [17] is the most popular and open source HSI data captured through aerial sensors. Data obtained from AVIRIS usually includes 200 useful bands and covers a wavelength of 400 up to 2500 nm. Reflective optics system imaging spectrometer (ROSIS) is another common example of airborne systems that covers a slightly narrower spectral range (430–860 nm) accumulating to 103 bands. The compact airborne spectrographic imager (CASI) includes 288 bands between 400 and 1000 nm [18]. Satellites in orbit are used for similar purposes and provide a broader view, however, with a tradeoff for a better resolution. Generally, satellites do not capture an extended range of spectral bands. For example, both Landsat-8 and Worldview-3 satellites capture eight bands only [19] while Sentinel-2 Satellite carries a multispectral instrument with 13 bands only. China's GaoFen-5 satellite stands out as the only one of its kind to capture HSI images with 330 bands, with a similar range to that of AVIRIS [20]. Other sensors include the NanoHyperSpec and short-wave infrared (SWIR) sensors that can be used to capture images from a ground platform or can be easily mounted on an unmanned aerial vehicle (UAV). NanoHyperSpec sensor can capture as many as 256 bands between 400 and 1000 nm. SWIR sensors vary based on the system design and can capture up to 430 bands at the higher end of the spectrum. Table II summarizes the most common HSI acquisition sources with more in-depth characteristics.

B. Common Tasks for HSI Using Machine Learning

This section describes the common use of machine learning models. These tasks include classification, segmentation, unmixing, super-resolution, and image translation. Each is briefly described below.

1) *Feature Extraction*: Feature extraction is an important step in processing hyperspectral images (HSI). This step involves transforming the raw high-dimensional spectral data into a lower dimensional space while retaining the most informative and discriminative features. This process simplifies the subsequent analysis and tasks, making them more computationally efficient and often more accurate. HSI data typically consists of hundreds of spectral bands, leading to high-dimensional datasets. Feature extraction reduces this dimensionality, thereby mitigating the curse of dimensionality. In addition, extracting relevant features can help filter out noisy and irrelevant information, ultimately improving the robustness of subsequent analysis. By focusing on key spectral features, the extracted data becomes more interpretable, and the underlying material easier to understand. Finally, effective feature extraction often leads to better performance in certain tasks, such as classification, as the extracted features are more discriminative.

Commonly used methods for feature analysis of HSI include principal component analysis (PCA) [21], [22], [23], linear discriminant analysis [24], [25], [26], [27], independent component analysis [28], [29], [30], and wavelet transform [31], [32]. However, more recently, deep learning techniques have been used for feature extraction including regular CNNs [33], [34], [35] or graph convolutional network [36], [37], [38]. In this

TABLE II
COMMON HSIS SENSORS SPECIFICATIONS

Imaging System	Platform	# Bands	Wavelength (nm)	Spectral Resolution (nm)	Spatial Resolution (m)
AVIRIS	Airborne	224	400 – 2500	9.6 – 10	20
ROSIS	Airborne	103	430 – 860	4	1 – 6
CASI	Airborne	288	400 – 1000	1.9 – 3.8	0.2 – 1.5
HYDICE	Airborne	210	400 – 2500	10	1 – 4
GaoFen-5	Satellite	330	400 – 2500	5 – 10	30
EO-1	Satellite	242	400 – 2500	10	30
HyperSpec	Ground/Airborne	256	400 – 1000	2.2	0.25 – 5
SWIR InGaAs	Ground/Airborne	430	900 – 2500	3.7	0.25 – 5

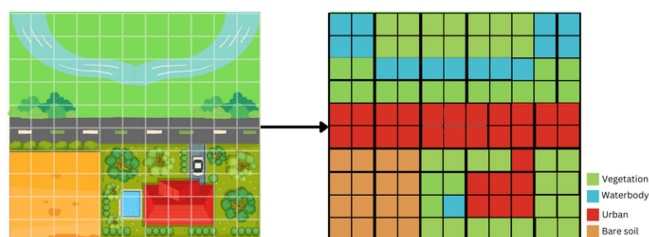


Fig. 1. Classification and segmentation of HSIs example where each pixel is assigned to a specific class.

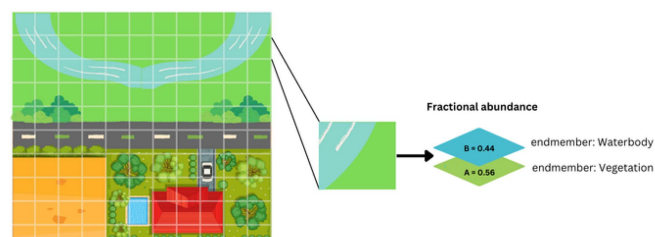


Fig. 2. Spectral unmixing of HSIs example showing the fractional abundances for the respective end members seen in a mixed pixel.

review, however, we focus on GNNs as they seem to result in better feature extraction results when compared to other neural network architectures, as illustrated later in this work.

2) *Classification and Segmentation*: Classification in the context of HSI is the process of categorizing pixels in an image into predefined classes based on their spectral signatures [39]. For example, given a hyperspectral image of a large agricultural area, different crops (e.g., corn, wheat, barley) can be classified into their respective groups. Segmentation, on the other hand, involves partitioning an HSI image into multiple segments or regions, each of which is homogeneous in certain properties based on spectral characteristics. Unlike classification, which assigns a class label to each pixel, segmentation groups pixels into meaningful regions based on similarities in their spectral data. For example, using HSI of an agricultural area, segmentation can delineate field boundaries and identify areas with different crop types or health statuses [40]. When using HSI, classification, and segmentation are typically the same task when the images are captured with a vertical 2-D view using an aerial vehicle. In such tasks, models attempt to assign a class to individual pixels. An example of image classification and segmentation is shown in Fig. 1. This task is mostly common in land use and land cover applications where pixels are classified based on the land use type (e.g., water body, vegetation, urban, etc). However, when the hyperspectral images are not aerial, classification and segmentation tasks become different. In such cases, each image usually contains a single class, where spectral similarities are used to classify images of the same class. Alternatively, an image may contain multiple objects, in which case spectral differences can be used to segment objects within a single scene.

3) *Spectral Unmixing*: The purpose of spectral unmixing is to decompose the spectral signature of each pixel into a collection of pure spectral signatures called endmembers, and their corresponding proportions called abundances [41]. Pure spectral signatures represent distinct materials or classes in the scene while the latter stands for the proportions of each end member present in a pixel. For example, hyperspectral images of farmlands often contain mixed pixels with varying amounts of soil, crops, and water. By unmixing the spectra, farmers can monitor crop health, soil moisture levels, and detect the presence of weeds or other materials. Usually, unmixing is based on well-established mathematical models like the fully constrained least squares [42], nonnegative matrix factorization [43], or iterative error analysis models [44]. Nevertheless, neural networks can be trained to learn the nonlinear relationships between observed spectra and end members. An example of spectral unmixing can be seen in Fig. 2.

4) *Resolution Enhancement*: Resolution enhancement in the context of HSI refers to improving the spatial resolution of hyperspectral images without sacrificing their spectral quality [45]. HSI data typically have high spectral resolution (e.g., many narrow spectral bands) but often suffer from relatively low spatial resolution compared to other imaging modalities, such as RGB or multispectral imaging. The objective of resolution enhancement is to enhance the spatial resolution of HSIs. An example of resolution enhancement can be seen in Fig. 3. For instance, when using HSI in the agricultural domain, enhanced resolution enables monitoring of crop health, identification of different crop types, detection of diseases at an early stage, differentiating between crops, weeds, and pests, allowing for targeted treatment,

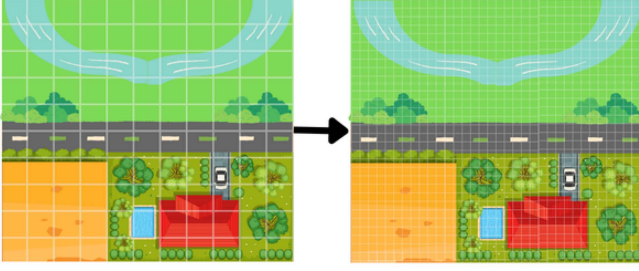


Fig. 3. Resolution Enhancement of HSIs example where the spatial resolution is enhanced to capture more detailed information at the pixel level.

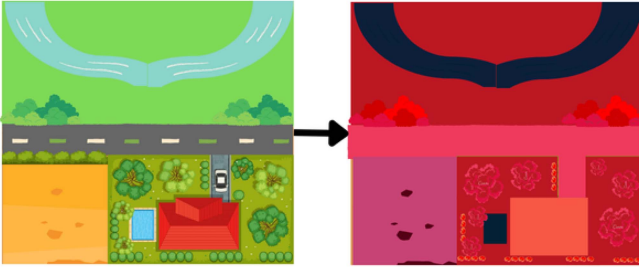


Fig. 4. Image translation of HSIs example where the image is translated to a different domain (NIR) with pseudocolors for illustration purposes.

and detailed mapping of soil properties, moisture levels, and nutrient distribution. Super-resolution techniques aim to generate high-resolution (HR) hyperspectral images from LR ones while preserving the spectral fidelity of the data [46]. This task is important as HSIs usually suffer from noise which hinders the accuracy of other tasks, such as classification or spectral unmixing.

5) *Data Synthesis and Translation*: Image translation refers to transforming an image from one domain to another, maintaining the original content while changing the style or appearance based on the target domain [47]. An example of image translation is shown in Fig. 4. Transforming hyperspectral data from one domain to another is usually used to enhance the spatial or spectral resolution, reduce noise, or adapt the image to a different sensing condition or modality [e.g., visual spectrum to near infrared (NIR)]. This process is critical for various applications in remote sensing, agriculture, environmental monitoring, and more where generative models can be useful.

C. Common HSI Datasets for Training

Hyperspectral data are difficult to obtain and only few open-source datasets are available for public use when compared to other domains. Generally, HSI datasets can be divided into three categories based on their original purpose. The majority of datasets were intended to be used for classification and segmentation tasks and are usually obtained through airborne mounted sensors. The datasets include the Pavia Center (PC) [48] and Pavia University (PU) [49] datasets, Indian Pines (IP) dataset [50], Kennedy Space Center (KSC) dataset [51], Salinas (SA) dataset [52], University of Houston (UH) dataset [53], Washington D.C. Mall (DC) dataset [54], Chikusei (CH)

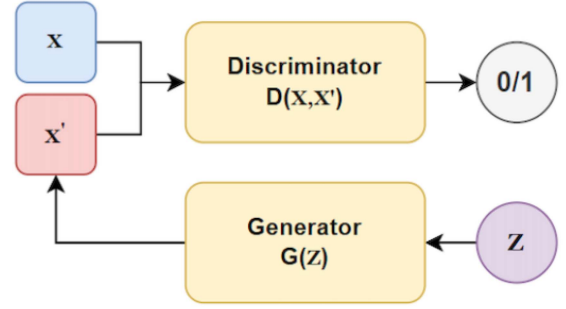


Fig. 5. Generative adversarial networks.

dataset [55], Xiongan Matiwan Village dataset [56], Eagle Reize dataset [57], HyRANK dataset [58], and finally, Houston dataset [59]. The second category of datasets is intended to be used for unmixing tasks where images are captured at LR and suffer from material mixtures within a single pixel [60], [61]. The most used datasets for unmixing tasks include the Simson dataset [62], Jasper Ridge dataset [63], and Urban dataset [64]. The third category of datasets includes Harvad [65], CAVE [66], NUS [67], ICVL [68], ARAD [69], and Berlin-Urban-Gradient dataset [70]. These datasets, which are less frequently used datasets, are typically leveraged for image translation or super-resolution tasks. It is worth mentioning that all available open-source datasets were captured over areas in North America and Europe apart from the CH dataset, which was captured over Japan in Asia. Lack of data from a diverse geographical area hinders accurately assessing scalability or generalizability of the various models. In addition, none of the aforementioned datasets provide recently captured images nor are they up-to-date with the technological advancements. Table III shows more detailed information about the HSI datasets discussed in this section.

D. Generative Models

Generative models are a class of machine learning models designed to generate new data instances [71]. Since the introduction of generative adversarial networks (GANs) [72], these models have been applied to various tasks, such as image synthesis, style transfer, and data augmentation. Large generative models designed for specific domains or data types are often referred to as *foundation models* [73]. Below, we briefly describe the most common types of generative models.

1) *Generative Adversarial Networks*: A GAN consists of two neural networks: a generator and a discriminator, trained simultaneously through an adversarial process, as shown in Fig. 5. The generator produces data samples (X') from random noise (Z) that mimic the real data (X), while the discriminator distinguishes between real and generated samples. This competitive process pushes the generator to produce increasingly realistic outputs. GANs have been extended with conditioning inputs, such as text or images, enabling targeted data generation.

2) *Energy-Based Models (EBMs)*: EBMs involve two main components, as illustrated in Fig. 6: a generator and an aligner. The generator transforms random noise (Z) into data-like outputs (X'), while the aligner maps parts of the input data to

TABLE III
COMMON HSI DATASETS AND SPECIFICATIONS

Dataset	Sensor	Category	# Useful Bands	# Scenes	Scene Size (px)	Location
CH	HyperSpec	Classification	128	77592	2517 × 2335	Ibaraki, Japan
IP	AVIRIS	Classification	200	10249	145 × 145	Indiana, U.S.A.
KSC	AVIRIS	Classification	176	4675	512 × 614	Florida, U.S.A.
PC	ROSIS	Classification	102	7456	1096 × 1096	Pavia, Italy
PU	ROSIS	Classification	103	42776	610 × 340	Pavia, Italy
SA	AVIRIS	Classification	200	54129	512 × 217	California, U.S.A.
UH	CASI	Classification	48	13511	349 × 1905	Texas, U.S.A.
DC	HYDICE	Classification	191	76237	256 × 256	Washington, U.S.A.
Xiongan Matiwan Village	Spectrometer	Classification	250	1	3750 × 1580	Matiwan Village, China
Eagle Reize	SPECIM AsiaEAGLE	Classification	247	1	714 × 714	Samford Valley, Australia
HyRANK	Hyperion	Classification	176	5	250 × 1376	Not Applicable
Houston	ITRES-CASI 1500	Classification	144	1	349 × 1905	Texas, USA.
Samson (SM)	AVIRIS	Unmixing	198	1	952 × 952	Entire U.S.A.
Jasper Ridge (JR)	AVIRIS	Unmixing	145	1	512 × 614	California, U.S.A.
Urban (UB)	HYDICE	Unmixing	162	1	307 × 307	Virginia, U.S.A.
Berlin-Urban-Gradient	Simulated EnMAP	Unmixing	111	3	817 × 220	Berlin, Germany
Harvard	Nuance Camera	Others	31	50	1392 × 1040	Massachusetts, U.S.A.
CAVE	CCD Camera	Others	31	32	1024 × 1024	Not Applicable
ICVL	DX4 Camera	Others	31	201	1392 × 1300	Not Applicable
NUS	Canon CRF	Others	49	66	1312 × 1924	Not Applicable
ARAD	IQ Camera	Others	31	510	482 × 512	Not Applicable

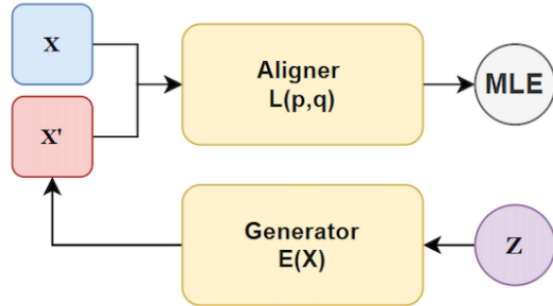


Fig. 6. Energy-based models.

specific energy states. maximum likelihood estimation is often used to assign lower energy values to more probable data configurations, making EBM suitable for tasks like density estimation and structured prediction.

3) *Autoencoders (AEs)*: AEs learn efficient data representations by compressing input data (X) into a lower dimensional latent space (Z) and reconstructing the input (X') from this compressed form, as depicted in Fig. 7. AEs are valuable in generative tasks due to their ability to capture essential features of the input data. Variational autoencoders (VAEs) extend AEs by learning a probabilistic latent space, allowing them to model data distributions more effectively.

4) *Flow-Based Models (FBMs)*: FBMs use invertible transformations to model complex data distributions, enabling exact likelihood computation and efficient sampling. These models follow a process similar to AEs, encoding the input into a latent

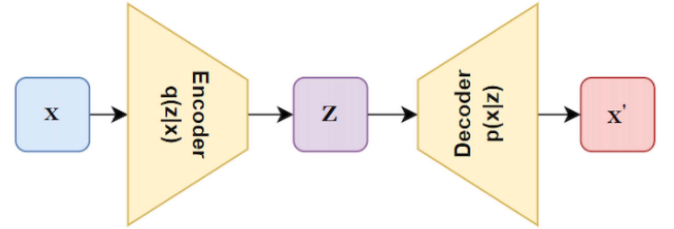


Fig. 7. Autoencoders.

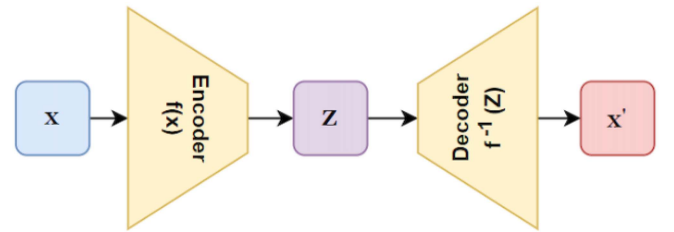


Fig. 8. Flow-based models.

space and reconstructing it from there, as shown in Fig. 8. FBMs are particularly effective for tasks requiring precise density estimation.

5) *Diffusion Models*: Diffusion models employ a forward process to gradually add noise to data and a learned reverse process to remove this noise, generating new samples. The forward diffusion process transforms data into pure noise over time, while the reverse process reconstructs the data distribution,

TABLE IV
SUMMARY OF GNNs MODELS CURRENT STATUS

Model Type	Advantages	Disadvantages
GANs	High-quality data generation; versatile for tasks like image synthesis and augmentation	Training instability, mode collapse, and extensive hyperparameter tuning required
EBMs	Flexible in capturing complex energy states; effective for density estimation	Requires careful parameter tuning; less commonly used compared to GANs or diffusion models
AEs	Efficient data compression and reconstruction; useful for dimensionality reduction	May produce lower quality samples compared to GANs or diffusion models
FBMs	Exact likelihood computation and efficient sampling	Less explored in practice; niche applications for density estimation
Diffusion Models	Stable training; generates highly realistic data with scalability	Computationally expensive; rapidly gaining attention for its performance

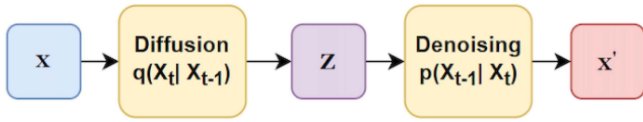


Fig. 9. Diffusion models.

as illustrated in Fig. 9. Diffusion models have shown remarkable performance in generating high-fidelity images and other complex data types.

Generative models have emerged as powerful tools in machine learning, each offering distinct advantages and facing specific limitations. GANs are renowned for producing high-quality data and their versatility in applications like image synthesis and data augmentation; however, they suffer from challenges, such as training instability, mode collapse, and the need for extensive hyperparameter tuning. EBMs are adept at capturing complex energy distributions, making them effective for density estimation, but they require meticulous parameter tuning and remain less popular than other approaches. AEs and their variational counterparts (VAEs) excel in efficient data compression, dimensionality reduction, and probabilistic modeling, though their generated outputs often lack the quality of GANs or diffusion models. FBMs provide exact likelihood computation and efficient sampling, making them valuable for density estimation, but their applications are limited to niche domains. Diffusion models, meanwhile, are gaining widespread attention for their stability and ability to generate highly realistic data, though they are computationally intensive. Collectively, these models have revolutionized generative modeling but require ongoing advancements to mitigate their drawbacks and broaden their applications. The advantages and drawbacks of the aforementioned models is highlighted in Table IV.

E. Evaluation Metrics

Evaluation metrics in hyperspectral image (HSI) vary depending on the task. In the following, we describe the metrics commonly used for classification, segmentation, spectral unmixing, resolution enhancement, and image translation tasks.

In HSI classification, the goal is to classify each pixel into a specific category based on its spectral signature, such as

identifying vegetation types or mineral compositions. Three major metrics are widely used for this task. Overall accuracy (OA) [74] measures the proportion of correctly classified pixels across the entire dataset and is defined as

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP represents true positives (correctly classified positive instances), TN represents true negatives (correctly classified negative instances), FP represents false positives (negative instances incorrectly classified as positive), and FN represents false negatives (positive instances incorrectly classified as negative).

Average accuracy (AA) [75] computes the average of the classification accuracies for each class, providing a balanced view of the performance between classes. It is given by

$$AA = \frac{1}{c} \sum_{i=1}^c \text{Accuracy}_i \quad (2)$$

where c is the total number of classes, and Accuracy_i represents the classification accuracy for the i th class.

The Kappa coefficient (K) [76] measures the agreement between the predicted and actual classifications, adjusted for the level of agreement that might occur by chance. It is calculated as

$$K = \frac{\text{Pr}(a) - \text{Pr}(e)}{1 - \text{Pr}(e)} \quad (3)$$

where $\text{Pr}(a)$ is the observed agreement (proportion of actual matches between prediction and ground truth) and $\text{Pr}(e)$ is the expected agreement by chance. The Kappa coefficient ranges from 0 (no agreement) to 1 (perfect agreement). For segmentation tasks, which aim to partition an image into distinct regions based on spectral properties, these metrics can also be applied.

Spectral unmixing involves decomposing each pixel into a combination of endmembers (pure spectral signatures) and their respective abundances. Key metrics include the abundance angle distance (AAD) [77], which measures the angular similarity between two abundance vectors. It is defined as

$$AAD(\mathbf{a}_1, \mathbf{a}_2) = \arccos \left(\frac{\mathbf{a}_1 \cdot \mathbf{a}_2}{\|\mathbf{a}_1\| \|\mathbf{a}_2\|} \right) \quad (4)$$

where $\mathbf{a}_1$ and $\mathbf{a}_2$ are abundance vectors (arrays representing the proportions of endmembers for a given pixel), and $\|\mathbf{a}\|$ denotes the Euclidean norm of vector $\mathbf{a}$.

The Abundance Fraction Error (AFE) [78] quantifies the average absolute error between true and estimated abundance vectors

$$\text{AFE} = \frac{1}{N} \sum_{i=1}^N |a_{\text{true},i} - a_{\text{est},i}| \quad (5)$$

where N is the number of pixels, $a_{\text{true},i}$ is the true abundance for pixel i , and $a_{\text{est},i}$ is the estimated abundance for pixel i .

Reconstruction error (RE) [79] evaluates the fidelity of reconstructed spectral data and is expressed as

$$\text{RE} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\| \quad (6)$$

where $\mathbf{x}_i$ is the original spectral vector for pixel i , and $\hat{\mathbf{x}}_i$ is the reconstructed spectral vector for the same pixel.

Pearson correlation coefficient (PCC) [80] assesses the linear correlation between two variables or vectors. It is calculated as

$$\text{PCC}(\mathbf{a}, \mathbf{b}) = \frac{\sum_{i=1}^N (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_{i=1}^N (a_i - \bar{a})^2} \sqrt{\sum_{i=1}^N (b_i - \bar{b})^2}} \quad (7)$$

where $\mathbf{a}$ and $\mathbf{b}$ are the vectors being compared, a_i and b_i are their individual elements, and $\bar{a}$ and $\bar{b}$ are their respective mean values.

Metrics for super-resolution tasks assess both spatial and spectral fidelity. Spectral angle mapper (SAM) [81] measures spectral similarity based on angular distance

$$\text{SAM} = \arccos \left(\frac{\langle u, v \rangle}{\|u\| \|v\|} \right) \quad (8)$$

where u and v are spectral vectors, $\langle u, v \rangle$ is their dot product, and $\|u\|$ and $\|v\|$ are their respective magnitudes.

Peak signal-to-noise ratio (PSNR) [82] quantifies the quality of a reconstructed image relative to a reference image and is calculated as

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{R^2}{\text{MSE}} \right) \quad (9)$$

where R is the maximum possible pixel value (e.g., 255 for 8-bit images) and MSE is the mean squared error between the reconstructed and reference images.

Structural similarity index measure (SSIM) [83] evaluates image quality based on luminance, contrast, and structural similarity. It is expressed as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (10)$$

where x and y are the images being compared, μ_x and μ_y are their mean luminance values, σ_x and σ_y are their standard deviations, σ_{xy} is their covariance, and C_1 and C_2 are small constants to prevent division by zero.

For image translation tasks, in addition to commonly used metrics, such as RMSE, SIM, ERGAS, and SSIM, spectral information divergence (SID) [84] quantifies the difference between

two spectral distributions. It is defined as

$$\text{SID}(p, q) = \frac{1}{2} [\text{KL}(p\|q) + \text{KL}(q\|p)] \quad (11)$$

where p and q are spectral distributions, and $\text{KL}(p\|q)$ is the Kullback–Leibler divergence between p and q .

III. GNN APPLICATIONS USING HSI

Hyperspectral imaging (HSI) has emerged as a critical tool for numerous machine learning applications, with significant advancements driven by the integration of GNNs. This article provides a comprehensive review of current developments in using generative models for three key HSI middle-stream tasks: feature extraction, image resolution enhancement, and data synthesis and translation. GNNs offer promising solutions for addressing challenges in these areas by leveraging their ability to generate high-quality spectral and spatial representations. This section highlights the trends and methodologies in HSI applications while also presenting insights from the surveyed studies.

The review was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) approach [86]. A total of 137 papers were initially reviewed, with 46 selected for inclusion based on strict eligibility criteria, including publication date (2019–2024), relevance to HSI tasks, and methodological contributions. Notably, a significant proportion (78.95%) of the reviewed works focused on national efforts, with China being the dominant contributor. The remaining 21.05% represented international collaborations, all of which included China as a key participant. Studies that heavily reduced HSI data into multispectral (MSI) or RGB formats [87], relied on spectral signatures [88], [89], or used Fourier transformations [90], [91], [92] were excluded. Papers that simply applied existing architectures without proposing innovative solutions were also not considered.

The detailed PRISMA methodology, including the number of surveyed and selected papers, eligibility criteria, and information sources, is summarized in Table V. By focusing on novelty, impact, and performance evaluation on benchmark datasets, this review highlights the potential of GNNs to advance HSI applications while identifying key challenges and research gaps.

A. Feature Extraction

Feature extraction is a critical stage in any machine learning project. An improved feature extraction strategy can significantly enhance the performance of an architecture. HSI poses a great challenge in this regard due to high dimensionality and large data volume. For example, many techniques like least squares methods [93], maximum likelihood, Random Forest (RF) [94], and K-nearest neighbors [95] have been proposed for feature extraction. However, these techniques struggle with the high dimensional nature of HSI [96]. Support vector machines (SVMs) is less sensitive to high dimensionality and was considered a good approach to use for the hyperspectral image

TABLE V
PRISMA METHOD QUESTIONS

Method	Description
Papers Surveyed	137 papers were surveyed overall.
Selected Papers	46 papers have been selected in total for this review.
Eligibility criteria	1) The paper should be published within the last five years (post 2019). 2) The work presented should not heavily reduce the HSI and turn them into MSI or RGB images. 3) The work should utilize HSI without adopting critical transformations such as Fourier transformation or spectral signatures. 4) The work presented should provide significant contribution and does not simply employ a known architecture to examine its applicability to a address a certain problem.
Information sources	MDPI, IEEEXplore, Taylor and Francis, Elsevier, ScienceDirect, ResearchGate, ACM, and arXiv were used to collect the related papers for this review.

classification task [97], [98], [99]. However, while SVMs outperformed traditional techniques, the extracted features tended to be shallow and this reflected negatively on the overall performance.

Deep learning models, such as convolutional neural networks (CNNs) [52], [100], [101], recurrent neural networks [102], [103], [104], hybrid methods with vision transformers (ViTs) [103], [105], and fusion networks (FNs) [106], [107], [108], [109], [110] have been proposed as alternatives to SVMs for classification tasks. However, these models are data-hungry and cannot be trained on a small number of samples without running into overfitting problems [96]. Therefore, many studies explored the potential of using generative models to extract relevant information from HSI. These studies assume that adversarial models will learn better discriminating features which can help boost the performance in different tasks including classification, segmentation, and object detection.

Generative approaches using variants of the diffusion model outperformed other architectures when learning spectral and spatial features. For example, Diff-HSI model [111] is the best-known performing architecture that utilized generative capabilities to perform feature extraction for classification tasks on HSI. Diff-HSI architecture used a pretrained denoising diffusion probabilistic model (DDPM) with randomly sampled and unlabeled patches preprocessed using PCA to learn effective spectral and spatial features. This research showed that features belonging to a similar class tended to cluster closely together as the time step increased, forming larger groups. Further, the authors introduced a dynamic feature fusion module to obtain multitime-step representations and further reduce the computational cost of training. The feature fusion module behaved similarly to the attention mechanism and was similar in design where important features were assigned higher weights enabling a balance between performance and memory footprint. Another diffusion-based approach is the subspace clustering model (DiffSC) [112],

which is similar in concept to Diff-HSI. However, the authors attempt to obtain the feature representations using the forward pass only at randomly selected timesteps. The primary intuition behind this approach was that the DDPMs purpose is not to generate images but rather to learn features. Hence, the inverse pass was not needed as similar feature maps could be directly extracted from the forward pass network. Finally, Spectral-Diff architecture [113] utilized the regular U-Net in the forward pass and utilized an AE for classification later on. The performance of this model was worse than other diffusion models which can perhaps be partially attributed to the unnecessarily aggressive down-sampling of the HSI.

GANs have also been used to extract better features for the classification task as well. Two representative examples of work in this area are the Residual-GANs [75] and the Two Branch (TB)-GANs [119], respectively. The Residual-GAN adds an 18-layer residual net to the discriminator model. The model includes two routes, one that implements a non-convolutional residual block (NCRB) and another is a convolutional block. Eventually, both propagation routes are added together at the last layer. The authors justify their design by claiming that the model would adopt the NCRB route when the input and output shapes are identical, and the other route would be used if they were not. Once the pretraining was completed, the learned feature maps were used as input to a classifier. The TB-GAN design, on the other hand, used a slightly different approach as two generators were used instead of one. The first generator was used for spectral features only while the second was for spatial features. The discriminator consisted of two paths for each given input. The raw representations were used to propagate through the spectral generated features and PCA was used to extract the corresponding spatial features. The performance of both models is between DiffSC and Diff-HSI models across multiple benchmark datasets. However, diffusion models were much faster than GAN models given their simpler architecture. For example, the DiffSC model is almost 5.7 times faster than TB-GAN while the performance of both models was about the same. In comparison, the SpectralMamba model [120], leveraging state-space modeling, offers a lightweight yet highly efficient architecture that obtained an OA of 91.88%, AA of 92.70%, and Kappa Coefficient of 91.19% on the Botswana HS Dataset while significantly reducing computational costs. SpectralMamba's selective state-space design, combined with its dynamic merging strategy, makes it a promising alternative to diffusion models for feature extraction.

Other attempts at extracting useful features include combination of multiscale residual attention with GANs (MSRA-G) [116], reducing HSI complexity by using an AE to create a latent space and train the GAN using the latent space (GLS) instead of the full-scale image [114], dimension reduction using dynamic mode decomposition (DMD) for domain adaptation [121], replacing the adversarial loss with a pixel content loss [122], and using a transformer-based discriminator model and up-scaling residuals to enhance feature extraction (TRUG) [117]. All these techniques are similar to the previously described efforts in terms of architecture and performance. Fig. 10 shows the different generative models' OA on a

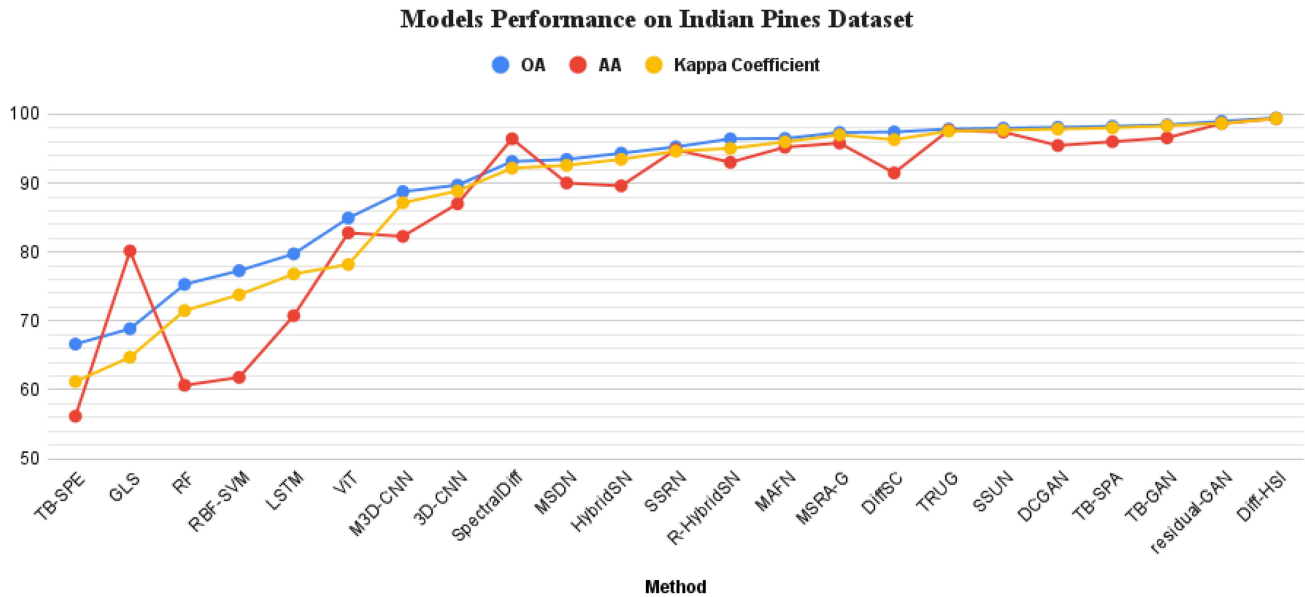


Fig. 10. Model performance using different feature extraction GNN methods on IP Dataset.

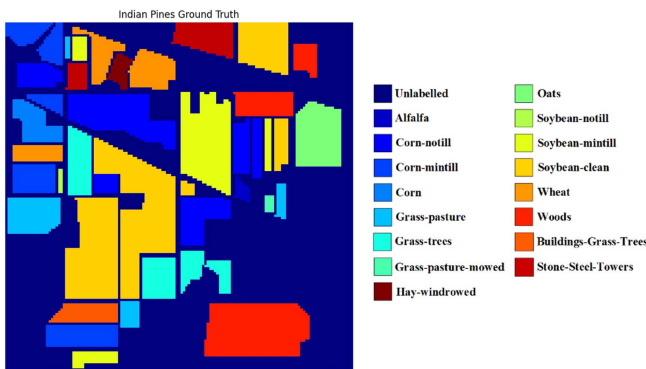


Fig. 11. Sample image taken from the IP Dataset [85].

common benchmark dataset—IP—seen in Fig. 11 in comparison to other models as well. Table VI summarizes feature extraction advancement in HSI using GNN while Table VII provides a more detailed comparison using all known evaluation metrics in the literature.

B. Image Enhancement

One key challenge in using HSI for machine learning is poor image resolution. There is always a tradeoff between spectral bands and spatial resolution. Common approaches to address poor resolution include using alternative representations like spatial-spectral sparse representations [128], high-order coupled tensor ring (TR) [129], and subpixel fusion using coupled spectral unmixing representations (CSU) [130]. Other methods include unsupervised adaptation learning (UAL) [131], transformers-based solutions [132], [133], subspace identification methods [134], and CNN-based approaches [135]. Generally, approaches that utilized fusion methods [136] tended to perform well. UAL approaches [131] were also promising due

to their unsupervised nature. There were also attempts to utilize GNNs to enhance the resolution of HSI without compromising spatial resolution. For example, Stable Super Resolution GAN [137] was used to create super-resolution images from HSI obtained through the GaoFen-5 Satellite. The proposed architecture is similar to the original SRGAN [138]; however, it replaced the residual blocks with bottleneck blocks in the generator and introduced a fully connected layer in the discriminator. While the results of this work are better than its counterparts, this research is limited to only the RGB bands for training the machine learning model and thereby the generated images were not hyperspectral. Attention enhanced GAN (AEGAN) [139] attempted to generate HSI by adding a spectral normalization layer to the generator and introducing different enhanced spatial attention modules (SAM). While this methodology was superior to other GAN-based approaches, the obtained results by diffusion models that followed a similar idea achieved better results. Diffusion models like the spectral diffusion prior (SDP) [141] aim to fuse LR HSI with a HR MSI. The architecture replaced the regular U-Net denoising network with an MLP-based network to avoid going downstream which is computationally expensive. One issue with SDP, however, is its sequential nature which does not provide sufficient flexibility for the network to capture relatively more important features in the HSI. HSR-Diff [142] attempted to address this problem by using conditional diffusion models. The primary contribution of this work was the introduction of conditional denoising transformers to tackle the loss of spectral information that occurs in vanilla transformers. These modifications made HSR-Diff the current state-of-the-art technique to generate super-resolution HSI with much better performance than that achieved by GANs [137], [139], and deep neural networks [143]. Fig. 12 shows the discussed approaches along with others' performance in terms of SAM Score on the common benchmark dataset (PU) shown in Fig. 13. Table VIII provides a detailed comparison of the various HSI approaches

TABLE VI
FEATURE EXTRACTION ADVANCEMENTS SUMMARY

Technique	Description
GLS [114]	Employs GLSs for HSIs classification.
3D-CSSEAN [106]	A 3-D cascaded feature extraction and attention based spectral-spatial element network.
Spectral-Diff [113]	A regular U-Net in the forward pass of a DDPM followed by an AE.
DMD [114]	Employs dimension reduction using DMD for domain adaptation.
SSRN [115]	SSRNet model proposes an efficient HSI and MSI three stages fusion architecture which employs a cross-mode message inserting operation, a reconstruction network, and spatial edge loss gaudiness.
R-HybridSN [52]	A joint spectral-spatial 3-D-CNN followed by spatial 2-D-CNN.
MAFN [109]	MAFN alleviate the influence of redundant bands and interfering pixels by utilizing a band attention module (BAM) and a SAM.
MSRA-G [116]	Combines multiscale residual attention with GANs to enhance HSI resolution at every band.
DiffSC [112]	Obtains the feature representations using the forward pass only at randomly selected timesteps.
TRUG [117]	Uses a transformer-based discriminator model and up-scaling residuals to enhance features extraction.
DCGAN [118]	An improved DCGAN architecture with double branches and transposed convolutions
TB-GAN [119]	Two generators are used instead of one. The first generator is used for spectral features only while the second is for spatial features.
Residual-GAN [75]	An 18-layer residual net is added to the discriminator model with two routes, one that implements a non-convolutional residual block (NCRB) and another which is convolutional.
Diff-HSI [111]	Use of a pre-trained DDPMs with randomly sampled and unlabeled patches that are pre-processed using PCA to effectively learn the spectral and spatial features

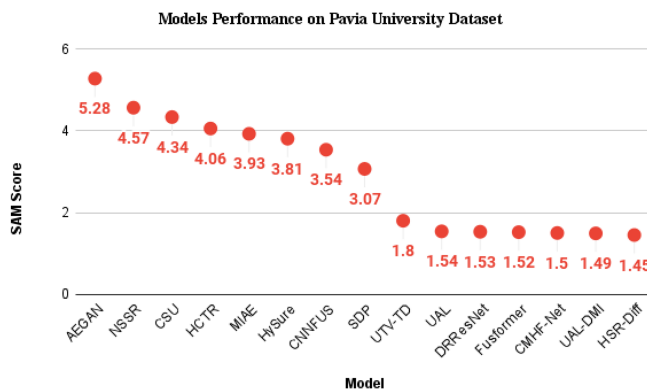


Fig. 12. Performance of different resolution enhancement generative models on PU Dataset by SAM Score.

TABLE VII
COMPARISON OF FEATURE EXTRACTION MODELS USING DIFFERENT EVALUATION METRICS

Technique	OA(%)	AA(%)	Kappa Coefficient(%)
TB-SPE [119]	66.65	56.18	61.21
GLS [114]	68.87	80.16	64.74
RF [123]	75.31	60.67	71.51
SVM [124]	72.36	83.16	68.88
LSTM [125]	79.74	70.77	76.82
ViT [124]	84.94	82.79	78.20
M3D-CNN [106]	88.77	82.28	87.15
3D-CNN [106]	89.71	87.01	88.87
Spectral-Diff [113]	93.15	96.49	92.18
MSDN [126]	93.44	90.02	92.59
HybridSN [52]	94.35	89.61	93.45
SSRN [115]	95.26	94.86	94.64
R-HybridSN [52]	96.43	93.02	95.04
MAFN [109]	96.48	95.22	95.99
MSRA-G [116]	97.35	95.81	97.00
DiffSC [112]	97.43	91.49	96.31
TRUG [117]	97.85	97.67	97.55
SSUN [127]	97.98	97.42	97.69
DCGAN [118]	98.11	95.46	97.85
TB-SPA [119]	98.26	96.02	98.02
TB-GAN [119]	98.45	96.58	98.33
Residual-GAN [75]	98.97	98.65	98.68
Diff-HSI [111]	99.46	99.37	99.38

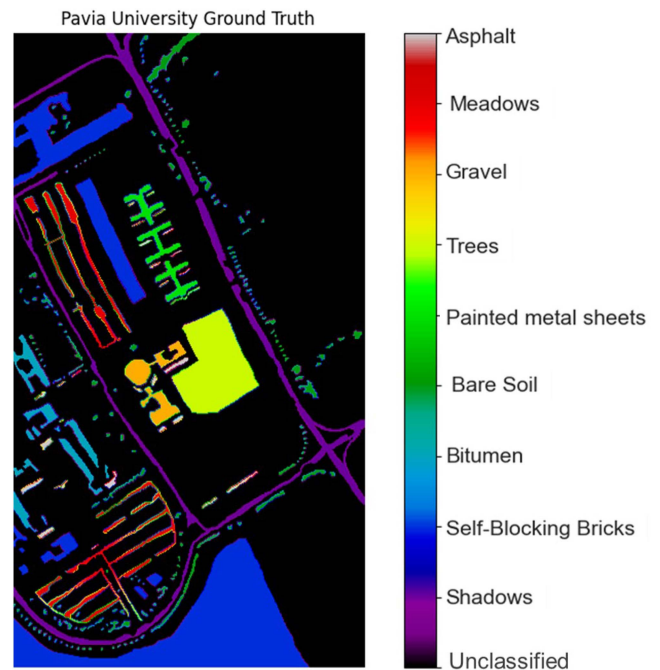


Fig. 13. Sample image taken from the PU Dataset [140].

TABLE VIII
RESOLUTION ENHANCEMENT ADVANCEMENTS SUMMARY

Technique	Description	SAM	PSNR	SSIM	ERGAS
AEGAN [139]	Uses attention-enhanced GAN to improve upon different SAM.	5.2823	46.5	0.9977	1.053
SSS [128]	Uses spatial-spectral-sparse representations of hyper-spectral images.	4.57	45.7	0.9966	1.113
CSU [130]	Utilizes pixel fusion using coupled spectral unmixing representations.	4.34	45.8	0.9965	1.096
HC-TR [144]	Uses tensor representation for hyperspectral images for feature extraction, data fusion, and resolution enhancement.	4.06	45.7	0.9965	1.109
MIAE [145]	A MIAE treats each pixel in a hyperspectral image as a sample that is fed into a super-resolution generative AE.	3.93	45.5	0.9965	1.111
HySure [146]	Uses HySure method to fuse the hyperspectral image with the panchromatic representation before fusing a multispectral image which has the same spectral coverage with the hyperspectral image.	3.81	45.4	0.9964	1.148
CNNFUS [135]	Utilizes significant correlation among spectral bands to approximate the super resolution HSI and utilize CNNs to denoise the approximation by utilizing alternating direction method of multipliers algorithm.	3.54	44.5	0.9952	1.236
SDP [141]	Fuses LR HSIs with a HR MSI and replaces the diffusion Unet with an MLP.	3.07	34.1	N/A	0.825
UTV-TD [147]	Employs TV and L1-norm to make use of HSI sparsity and piecewise smoothness alongside with classical Tucker decomposition for representation as the multiple of a sparse core tensor by dictionary metrics along the three modes	1.8	32.4	N/A	0.981
UAL [131]	Uses unsupervised adaption learning for resolution enhancement.	1.54	32.1	N/A	1.029
BRResNet [148]	Uses BAM for efficient feature extraction of hyperspectral images and utilizes that for resolution enhancement purposes.				
FusFormer [132]	A transformer-based architecture that relies on self-attention to extract intrinsic relationships.	1.52	32.1	N/A	1.045
CMHF-Net [149]	Introduces a model of better interpretability for MSI and HSI fusion and caters to cases where spatial and spectral mismatches occur in training and test data.	1.5	30.0	N/A	1.217
HSR-Diff [142]	Introduces conditional denoising transformers use to tackle the loss of spectral information.	1.45	29.5	N/A	6.6958

for image enhancement using relevant evaluation metrics available in the literature.

C. Data Synthesis and Translation

Generating synthetic hyperspectral data is important because HSI are scarce and difficult to obtain. Many approaches have been proposed to utilize GNNs to generate synthetic HSI. However, these attempts faced challenges due to the instability of GNNs given the high dimensionality of the data. For example, an improved version of DCGAN [118] was used to generate synthetic HSI spectral signatures. The use of double branches and transposed convolutions in this architecture enhances its performance. However, DCGAN reached Nash equilibrium as the generator overtook the discriminator quickly. Similarly, Palsson et al. [155] attempted to synthesize HSI using the standard GAN architecture with a modified spectral angle distance objective function. This model outperformed other generative approaches, including AE-based methods, such as CN-NAEU [156], legacy sparsity-constrained NMF methods [157], and unsupervised methods such that featured in [158]. HSI can

also be synthesized from RGB images. However, translation and reconstruction attempts to map RGB images to HSIs is difficult because RGB represents only three bands of the spectrum, which is not informative enough to generate a corresponding HSI that spans over a wider spectrum. HSGAN [159] addressed this issue by using a spatial-spectral attention block at every level of the generator. An attention block would be useful to allow better generalization across different RGB images and would also attempt to reduce the effect of noise that usually exists in real-world images. Their approach is better than other attempts at the same task, such as HRNet [69], EffCNN [160], and LSS [161], but the results can vary based on the dataset used. It is worth noting that evaluation metrics for mapping tasks are designed primarily for RGB images and it is unclear if these metrics fairly evaluate HSI reconstruction tasks. Another notable work in generating HSI is the hyperspectral and multispectral remote sensing image fusion using adaptive multi-perceptual implicit guided sampling (AMGS) GANs [153]. The primary idea behind AMGS is to utilize multispectral features to guide the model toward better spatial details. The results show that the AMGSGAN approach outperformed other synthetic data

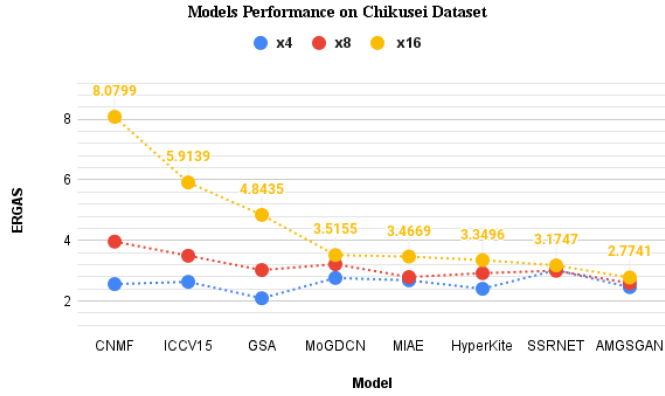


Fig. 14. Performance of different data synthesis generative models on Chikusi Dataset at three levels of data synthesizing.

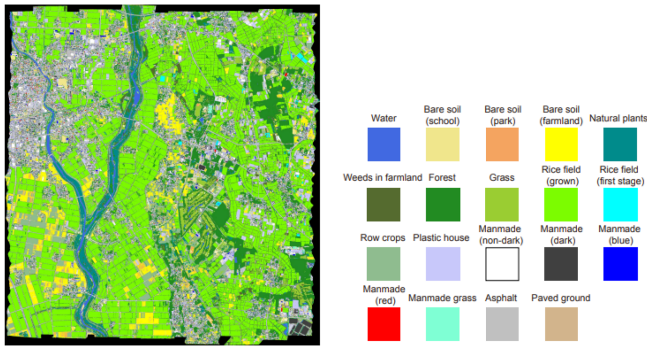


Fig. 15. Sample image taken from the CH Dataset [154].

generation techniques like HyperKite [48], MoGDCN [152], SSRNET [115], and model inspired autoencoder (MIAE) [145] especially as the number of synthetic data increases. Building on these advancements, HSiGene [14] introduced a groundbreaking approach to hyperspectral data synthesis using LDMs. A key feature of HSiGene is its multicondition controllable generation framework, which provides precise control over both the spectral and spatial characteristics of the synthesized HSIs. In addition, the model incorporates a two-stage super-resolution framework, where high-quality HSIs are generated using HR RGB images as guidance. This ensures fidelity in both spatial and spectral domains. Fig. 14 shows a comparison between different synthetic data generation techniques on the benchmark dataset of CH (seen in Fig. 15) in terms of the ERGAS metric at three different levels of synthetic data. Table IX summarizes advancements in data synthesis for HSI using GNNs, while Table X lists available open-source contributions of the techniques discussed earlier, enabling further exploration and reproducibility.

IV. DISCUSSION

GNNs hold immense potential in supporting various machine learning tasks for hyperspectral imaging (HSI), including feature extraction, resolution enhancement, and data synthesis. Despite these advancements, the practical deployment of generative

TABLE IX
DATA SYNTHESIS AND TRANSLATION ADVANCEMENTS SUMMARY

Technique	Description
CNMF [150]	Uses Coupled nonnegative matrix factorization (CNMF) to fuse hyperspectral images with multispectral ones, enabling better spatial resolution for all hyperspectral bands with minimal spectral distortion.
ICCV15 [151]	Utilizes a decomposition of HSI and MSI images to extract pure reflectance spectra and their respective mixing coefficients.
MoGDCN [152]	Conforms an iterative HSI super-resolution algorithm into a network and exploits the observation matrix for improved generalizability via subnetworks.
MIAE [145]	A MIAE treats each pixel in a hyperspectral image as a sample, feeding it into a generative model to create synthetic samples.
HyperKite [48]	Utilizes low-level feature extraction through bilinear interpolation and downsampling, with skip connections to bridge the gap between the reference and DIP-upscaled image.
SSRNET [115]	Fuses HSI and MSI through a three-stage network, producing a preliminary fused image and reconstructing spatial and spectral information from the HSI and MSI, respectively.
AMGSGAN [153]	Uses adaptive multiperceptual implicit guided sampling GANs, where multispectral features guide the model toward better spatial details.
HSiGene [14]	Uses LDMs for hyperspectral data synthesis with a multicondition controllable framework. Features a two-stage super-resolution process guided by HR RGB images, ensuring spectral and spatial fidelity

methods in HSI still faces challenges, requiring further investigation and refinement.

For feature extraction, models, such as GANs and DDPMs, have demonstrated remarkable performance in classification, segmentation, and object detection. While GANs excel in generating diverse and high-quality feature maps, diffusion models offer notable advantages in computational efficiency, training stability, and testing speed, as summarized in Table XI based on the numbers reported in each work and given the same dataset. These characteristics make GANs suitable for tasks requiring diversity in generated features, whereas diffusion models are preferable for scenarios with constrained computational resources or large-scale processing requirements. However, the limited availability of geographically diverse datasets continues to hinder the generalization of these models. Developing robust transfer learning frameworks and data augmentation techniques could address this issue, enabling more universal applications of feature extraction methods.

In resolution enhancement tasks, integrating transformers with attention mechanisms has consistently led to improved results across different generative techniques. Diffusion models, in particular, stand out for their temporal efficiency and scalability, making them an ideal choice for real-time applications or scenarios demanding high-throughput data processing. Nonetheless,

TABLE X
SUMMARY OF REVIEWED OPEN-SOURCE CONTRIBUTIONS

Technique	GitHub Repository
HyperKite [48]	https://github.com/wgcban/DIP-HyperKite.git
HybridSN [52]	https://github.com/gokriznastic/HybridSN
MAFN [109]	https://github.com/Li-ZK/MAFN-2021
Diff-HSI [111]	https://github.com/zjyaccount/MTMSD
Spectral-Diff [113]	https://github.com/chenning0115/SpectralDiff
SSRNET [115]	https://github.com/hw2hwei/SSRNET
UAL [131]	https://github.com/JiangtaoNie/UAL
FusFormer [132]	https://github.com/J-FHu/Fusformer
SDP [141]	https://github.com/liuofficial/SDP
MIAE [145]	https://github.com/liuofficial
UTV-TD [147]	https://liangjiandeng.github.io
BRResNet [148]	https://liangjiandeng.github.io/Projects_Res/bam_mm2021.html
AMGSGAN [153]	https://github.com/chunyuzhu/AMGSGAN
HSIGene [14]	https://github.com/LiPang/HSIGene

TABLE XI
COMPARISON OF INFERENCE TIME CARRIED ON RTX 3090 GPU ACROSS TECHNIQUES

Technique	Training Time (ms)	Testing Time (ms)
TB-SPE [119]	683.82	1.36
RF [123]	15.24	0.32
LSTM [125]	23.45	0.8
M3D-CNN [106]	206.56	1.99
3D-CNN [106]	11.35	25.48
SpectralDiff [113]	N/A	48.53
MSDN [126]	10.28	21.48
HybridSN [52]	6.03	5.8
SSRN [115]	9.58	15.48
R-HybridSN [52]	8.69	22.15
MAFN [109]	12.16	21.12
MSRA-G [116]	9.93	20.75
DiffSC [112]	221.92	N/A
SSUN [127]	54.52	2.17
DCGAN [118]	764.82	2.78
TB-SPA [119]	673.22	2.37
TB-GAN [119]	1262.02	2.81

these methods remain computationally intensive, especially for super-resolution, where the preservation of spectral and spatial fidelity is critical. Future research should prioritize the design of lightweight generative models that balance computational demands with accuracy, ensuring their feasibility for broader adoption in operational settings.

Data synthesis tasks, including image translation and augmentation, have witnessed rapid advancements through the use of attention-augmented GNNs. These methods ensure the

preservation of critical spectral and spatial features, enabling highly accurate reconstructions and augmentations. However, the evaluation metrics currently employed for assessing performance often fail to account for the unique complexities of HSI data. Metrics designed for RGB images inadequately reflect the spectral and spatial intricacies inherent to hyperspectral imagery. The development of hyperspectral-specific metrics is therefore essential to enable fair and accurate evaluation of generative methods in this domain.

The challenges posed by noise mitigation and dead pixels remain significant obstacles in hyperspectral imagery. Current solutions, such as spike removal through nearest-pixel analysis or density-based identification, are effective but computationally expensive and dataset-specific. Generative approaches, including AEs and regressive models, have proven successful in handling similar issues in simpler visual data. Their adaptation to HSI could offer flexible and resource-efficient alternatives, particularly for reconstructing corrupted data or addressing calibration errors. For example, AEs could leverage latent representations to effectively reconstruct missing data, while regression-based methods could predict and correct anomalies by learning spectral patterns. These adaptations would not only reduce preprocessing burdens but also improve the robustness of HSI applications in real-world environments.

Looking ahead, future research must address several pressing gaps. Developing computationally efficient generative techniques, particularly for tasks, such as super-resolution and noise mitigation, is critical. This includes exploring methods that optimize resource usage while maintaining or enhancing model performance. In addition, hyperspectral-specific evaluation metrics must be designed and validated to accurately assess generative methods, considering the dense and high-dimensional nature of HSI data. Emerging vision-language models, such as SpectralGPT, S2MAE, and HyperSigma, present promising avenues for universal spectral data processing. While these models demonstrate potential, their effectiveness compared to task-specific approaches remains uncertain. Comprehensive comparative studies are required to determine their relative strengths and weaknesses, offering insights into their practical applicability.

Training vision-language models requires substantial computational resources, while the development of GNNs often necessitates reducing computational expenses to enhance accessibility and scalability. Balancing these demands is essential to ensure practical adoption. A potential approach is leveraging pretrained vision-language models for HSI applications, fine-tuning only the essential layers for task-specific needs. This reduces the computational burden compared to end-to-end training. Furthermore, lightweight GNN architectures, optimized through techniques, such as pruning, quantization, and knowledge distillation, can reduce model complexity while retaining performance. The integration of modular training strategies may further bridge this gap by separating tasks into smaller, interdependent modules. For instance, pretrained generative models can handle feature extraction, with smaller networks trained specifically for resolution enhancement or noise mitigation. Employing data-efficient learning strategies, such as curriculum learning or active learning, can minimize the number of required training samples,

focusing computational resources on the most informative data. Lastly, advancements in hardware acceleration, such as the use of GPUs, TPUs, and distributed computing frameworks, can offset the high computational costs of vision-language model training. Combined with software optimizations like mixed-precision training and efficient deep learning libraries, these strategies can strike a balance between performance and resource constraints. Future research should explore the synergy between vision-language models and optimized GNN techniques to address the challenges of computational efficiency while advancing HSI applications. In addition, research must focus on bridging the gap between theoretical advancements and real-world implementations. Addressing the computational and data-related challenges will enable more robust and scalable applications, fostering progress in hyperspectral imaging and its associated domains.

V. CONCLUSION

GNNs offer significant potential for advancing hyperspectral imaging (HSI) applications by addressing challenges in classification, segmentation, and object detection. Techniques, such as GANs and DDPMs have shown superior performance in generating high-quality feature maps, with diffusion models offering additional advantages in computational efficiency and training stability. Similarly, transformers with attention mechanisms have demonstrated improved performance across generative methods, particularly in tasks, such as image translation, data augmentation, and synthesis, where preserving spectral-spatial correlations is critical. However, these methods also face limitations, including issues like mode collapse in GANs and resource-intensive implementations in diffusion models for large-scale applications. In addition, the reliability of these approaches depends on the quality and diversity of HSI datasets, which remains a persistent challenge in the field.

Future research should focus on developing more computationally efficient generative techniques, particularly for tasks like super-resolution and data synthesis, and on designing more accurate evaluation metrics that consider the complex spectral and spatial characteristics of HSI data. Furthermore, the integration of vision-language models provides an opportunity to explore the tradeoffs between task-specific and universal solutions, which could further enhance the scalability and versatility of HSI applications.

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Diaa Addeen Abuhani (Member, IEEE) received the B.Sc. degree in computer engineering, in 2023, from the American University of Sharjah, Sharjah, UAE, with a double minor in computer science and data science where he is currently working toward the M.Sc. degree in machine learning.

He is a Research Fellow with the MIT Senseable City Lab, Cambridge, MA, USA. His research interests include remote sensing for precision agriculture applications, generative neural networks for hyperspectral imagery, models optimization, and dimensionality reduction.



Imran Zuolkernan (Member, IEEE) received the B.S. (Hons.) and Ph.D. degrees in computer science from the University of Minnesota, Minneapolis, USA, in 1983 and 1991, respectively.

From 1992 to 1995, he was an Assistant Professor with the Computer and Electrical Engineering Department, Pennsylvania State University, Pennsylvania, USA, from 1995 to 1998 a Principal Design Engineer with AMCS Inc., Chanhassen, MN, USA, from 1998 to 2000, the Chief Executive Officer with Askari Information Systems, Pakistan, and the Chief Technology Officer with Knowledge Platform, Inc., Singapore. In 2003, he joined the American University of Sharjah, Sharjah, UAE, where he is currently a Professor and Head of Computer Science and Engineering. He has authored or coauthored more than 230 peer-reviewed articles.



Raghad Aldamani received the B.Sc. in computer science from the University of Sharjah, Sharjah, UAE, in 2023. She is currently working toward the M.Sc degree in machine learning with the American University of Sharjah, Sharjah.

She is a Visiting Student with the MIT Senseable City Lab, Cambridge, MA, USA. Her research interests include understanding urban city dynamics with respect to spatiotemporal changes, land-use, land-cover monitoring using remotely sensed satellite data, and models optimization and deployment for on-edge applications.



Mohamed Alshafai (Member, IEEE) is currently working toward the B.Sc. degree in computer science with a minor in data science with the American University of Sharjah, Sharjah, UAE.

His research interests include autonomous robotics in low gravity setups, generative neural networks for natural language processing with focus on Large Language Models (LLM)s, and algorithms complexity analysis and optimization.